A Novel DDS Using Nonlinear ROM Addressing With Improved Compression Ratio and Quantization Noise

Lakshmi S. Jyothi Chimakurthy, Malinky Ghosh, Fa Foster Dai, ,
and Richard C. Jaeger,

***Abstract*—This paper presents a novel direct digital frequency synthesis (DDFS) ROM compression technique based on two properties of a sine function: (a) piecewise linear technique to approximate a sinusoid, and (b) variation in the slope of the sinusoid at different phase angles. In the proposed DDFS architecture the ROM stores a few of the sinusoidal values, and the interpolation points between the successive stored values are calculated using linear and nonlinear addressing schemes. The nonlinear addressing scheme is used to adaptively vary the number of interpolation points as the slope of the sinusoid changes, leading to a greatly reduced ROM size. The proposed architecture achieves a high compression ratio with a spurious response comparable to that of recent ROM compression techniques. To validate the proposed DDS architecture, the linear, nonlinear, and conventional DDS ROM architectures were implemented in a Xilinx Spartan II FPGA and their spurious performances were compared.**

I. Introduction

Direct digital frequency synthesizers (DDFS) provide fast switching and precise frequency resolution. The simplest architecture of the DDFS uses an N-bit accumulator, a read-only memory (ROM), and a digital-to-analog converter (DAC), implemented using the same reference clock f_{clk} as illustrated in Fig. 1. The output frequency of the DDFS depends on the N-bit input word called the frequency control word (FCW).

At the edge of the reference clock, the FCW is added to the value in the N-bit accumulator. At any instant the value in the accumulator represents the phase of the sinusoid, so it is referred to as the phase accumulator register. The output of the phase accumulator register is used to address the ROM that stores the sinusoid sample values. The output frequency of the DDFS depends on f_{clk} as

$$f_{out} = f_{clk}\frac{FCW}{2^N}. \tag{1}$$

The finest frequency resolution possible is given by

$$f_{res} = \frac{f_{clk}}{2^N}. \tag{2}$$

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The authors are with the Department of Electrical and Computer Engineering, Auburn University, Auburn, AL 36849-5201 (e-mail: daifa01@eng.auburn.edu).

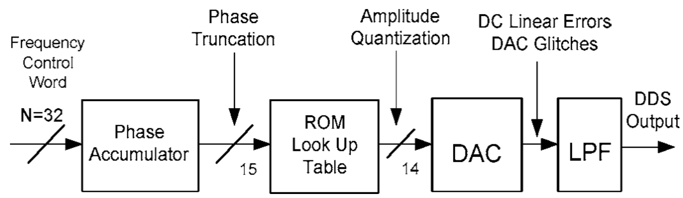

Fig. 1. Basic DDFS architecture.

The size of the ROM varies with the number of phase bits N as 2^N; therefore, to keep an affordable ROM size, the phase bits are often truncated. The phase truncation maintains the frequency resolution, yet generates spurious tones at the DDS output. The finite resolution of the ROM also causes quantization errors leading to a spurious output response. Therefore, several compression techniques have been proposed to optimize the ROM size and the output dynamic range. The output sine wave can be modeled by (3),

$$y = \sin\left(\frac{\pi x}{2}\right) + \varepsilon(x), \tag{3}$$

where x represents the output phase bit at every clock cycle and $\varepsilon(x)$ models errors such as amplitude resolution errors and algorithmic errors. The most common DDFS architecture was proposed by Tierney [1]. This architecture, however, incorporates a huge ROM. The conventional DDFS architecture utilizes a ROM look-up table to convert the accumulated phase word into sine/cosine words that are further used to drive a linear binary-weighted DAC. The speed bottleneck of a conventional DDFS architecture lies in the large look-up table with multilevel decoders. The huge look-up table not only restricts high-speed operation, but also occupies a large area and consumes a large amount of power. ROM compression techniques gain advantages such as increased speed and reduced die size, cost, power consumption, and digital switching noise. Hence, reduction of the ROM size using compression techniques is important for both high-speed and low-power applications.

To reduce the power dissipation and the die size, a nonlinear DAC with sine-weighted current cells can be employed [2]. The nonlinear DAC converts the digital phase information directly into an analog output. The sine function approximation algorithm presented in this paper can also be used in nonlinear sine-weighted DAC designs, even

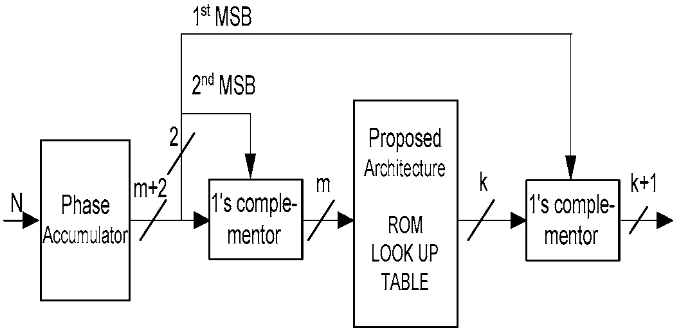

Fig. 2. Architecture for constructing sine function using symmetry around 2 and .

if no look-up table is needed in a ROM-less DDFS architecture.

This paper presents a compression technique that exploits the variation of slope of the sinusoid to vary the number of values that have to be interpolated among the values stored in the ROM [3]. In addition, the sine wave has been assumed to be piecewise linear among the successive stored values. A small multiplier along with some control circuitry is used to calculate the interpolation values. The results show that the technique can achieve a considerable reduction in ROM size with reduced circuit complexity without degradation in the spurious response, and presents no bottleneck to low-power applications.

Section II of this paper presents the generic algorithm for the proposed ROM compression technique. In Section II-B the linear addressing scheme of the architecture is presented, followed by the nonlinear addressing scheme of the architecture in Section II-C. The simulation results are presented in Section III. Section IV includes a detailed comparison of the present architecture with the various previous ROM compression techniques. This section also reviews the pros and cons of the prior art techniques. The architecture is validated by implementation on Xilinx Spartan II FPGA (San Jose, CA) and a measurement is presented in Section V along with brief analysis. The paper is concluded by Section VI.

A DDFS with phase resolution of 15 bits and amplitude resolution of 14 bits is discussed. A traditional DDFS implementation using the sine function quadrature symmetry compression technique, as shown in Fig. 2, requires a $2^{13} \times 14$-bit ROM [4]. A 15-bit phase word should result in -92.06 dBc of rejection in the magnitude of the worst-case spur due to truncation of the phase accumulator register. The proposed architecture [5] with nonlinear addressing scheme meets the performance goal of -90 dBc of spurious rejection with a ROM of 1216 bits, resulting in a compression ratio of 94.3:1 when compared to the abovementioned traditional DDFS implementation, whereas the algorithm of Nicholas [6], [7] could achieve a compression ratio of only 34.4:1 for the same specifications and performance. These compression ratios are compared to the traditional memory with quadrature symmetry technique implemented.

II. Proposed ROM Compression Algorithm

In the architectures of DDFS using ROM compression techniques, certain samples of the sinusoid are stored in the ROM, and the entire sinusoid is created by calculating the interpolation values using these stored values. In our architecture, the two most popular techniques of storage reduction, (i) sine function quadrature symmetry and (ii) sine-phase difference algorithm, have been used. The sine-phase difference algorithm technique stores the value of $\sin(x) - x$ instead of the value of the corresponding sine sample. An adder is used later to add the phase word to the value stored in the ROM. This technique helps to reduce by two bits the width of the word that is stored in the ROM. Incorporating this technique in our architecture requires two additional adders.

At any clock cycle, the value in the phase register indicates the phase of the sine function. As not all of the samples of the sinusoid are stored, only the first m bits of the N-bit phase word are used for addressing the ROM. The remaining d bits ($d = N - m$) are used to calculate the value and position of the interpolation point. The first-order linear interpolation formula is given as

$$\sin(B) = \sin(A) + (\sin(C) - \sin(A)) \times \Delta x, \tag{4}$$

where $\sin(A)$ and $\sin(C)$ are the successive stored sinusoid samples; $\sin(B)$ is the sample to be interpolated and is calculated using (4); $\sin(A)$ is the sine value addressed by the phase bits in the present clock cycle, and $\sin(C)$ is the next ROM sample. The value of Δx is obtained using d least significant bits (LSBs) of the phase word. There can be a maximum of 2^d interpolation points between the successive stored values of the ROM. The maximum number of samples is interpolated when the input FCW is 1.

The value in the phase registers at any instant reads

$$R = FCW \times (i - 1), \tag{5}$$

where i is the number of clock cycles, and $R[(N-1):0]$ represents an N-bit word with bit $N-1$ as its most significant bit (MSB) and bit 0 as its LSB. The notation is used in Verilog hardware description language. The phase angle of every sinusoid sample to be interpolated is calculated by (6) and the interpolated value is given by (7).

$$X_i = B = \left(\frac{\pi}{2}\right)\left(\frac{R_i[N-1:0]}{2^N}\right) \tag{6}$$

$$\sin(B) = \sin(A) + \frac{(\sin C - \sin A)}{(2^d)} \times (R[d-1:0]). \tag{7}$$

In other words, the d LSBs of the phase register output R indicate the relative position of the interpolated sample from the preceding stored sample. They therefore also represent the number of clock cycles required for the value in the phase register to change from A to B. The output of the ROM has a word length and hence amplitude resolution of k bits.

In the proposed architecture, the sine function is assumed to be piecewise linear. In order to avoid the division in (7), which involves hardware, the formula has been modified to

$$\sin(B) = (\sin(C) - \sin(A)) \times (R[d-1:0]) + \sin(A) \ll (d). \tag{8}$$

Multiplication of both sides by 2^d makes the output word length $n = k + d$ bits. This increases the precision of representation. The $k + d$ bits are later truncated to k bits to meet the specifications of the DAC. The spurious response is analyzed after truncating the word length to k bits. The memory requirements here are $2^m \times k$ bits, where m is the number of bits of the phase register output used for addressing the ROM and k is the ROM wordlength.

Memory requirements are further reduced by exploiting the variation in the slope of a sinusoid at every phase angle. The slope of the sinusoid changes from 1 to 0 as the angle changes from 0° to 90°. Thus, the slope of a sinusoid is high for smaller phase angles, gradually reducing as the phase angle approaches 90° and becomes 0 at 90°.

As a higher slope means a larger difference in successive sample values, the proposed architecture with a nonlinear addressing scheme stores more samples in the ROM lookup table in the region of the sinusoid having higher slope and a smaller number of samples in the region of the sinusoid with lesser slope. Thus, as seen in Fig. 3, the number of samples stored in the 0°–45° region is 64, as opposed to 32 samples stored in the 45°–90° region. Therefore, the number of interpolation values that are calculated is less in the high-slope region than in the low-slope region of the sinusoid. Though many such partitions of the sinusoid can be made where each part can have a different number of stored interpolation samples, for the architecture presented in this paper the sinusoid is partitioned into two parts: 0°–45° and 45°–90°. As shown in Fig. 3, the sine values for phases between 0° and 45° are stored in ROM1, and those for 45°–90° in ROM2. The number of ROM values stored in the ROM2 can be reduced to almost half without a significant drop in the spurious response.

As there are two ROMs, the number of addressing lines to ROM1 is taken as m and, accordingly, the number of addressing lines to ROM2 is denoted as $m - 1$. Thus, the number of interpolation values that are calculated between 45° and 90° increases to twice the number that are calculated for the single ROM architecture with a linear addressing scheme, and their relative position is indicated by LSB d_2-bits ($d_2 = d+1 = N-m+1$) bits. For the above explained nonlinear addressing scheme, (8) is modified to

$$\sin(B)_2 = (\sin(C)_2 - \sin(A)_2) \times (R[d_2-1:0]) + \sin(A)_2 \ll (d_2), \tag{9}$$

$$\sin(B)_1 = (\sin(C)_1 - \sin(A)_1) \times (R[d_1-1:0]) + \sin(A)_1 \ll (d_1). \tag{10}$$

Subscripts 2 and 1 indicate that the values under discussion are from ROM2 and ROM1, respectively. d_2 is one bit greater than d_1 ($d_1 = d = N - m$). If the widths of both ROM1 and ROM2 are taken as k bits, as in the linear addressing scheme, shifting the $\sin(A)_2$ value by d_2, as given in (9), will make the output word length $k+d+1$ bits. In order to have uniform output, the width of ROM2 is made $k-1$ bits instead of k bits. The memory requirements for this architecture are $2^{m-1} \times k + 2^{m-2} \times (k-1)$ bits.

[illegible]

Without losing generality, a phase precision of 15 bits and a ROM of 2^8 bits with an amplitude resolution of 14 bits are used to simulate the proposed architecture as shown in Fig. 4. The width of ROM is reduced from 14 bits to 12 bits due to incorporation of the sine-phase difference algorithm. The block marked as the sine-phase difference algorithm adds the value of phase angle A back to the output of the ROM ($\sin A - A$) such that the output of this block is $\sin A$ value with 14-bit resolution. Thus the actual ROM size is $2^8 \times 12$ bits.

As a ROM of length 2^8 has been used, 256 sinusoid samples for phase angles of 0°–90° are stored in the ROM. For a phase accumulator of length 15 bits, the phase accumulator output word is represented as $R[14:0]$ where bit 14 is MSB and bit 0 is LSB. The first two of the 15 bits of the phase accumulator register indicate the quadrant of the sine wave. Hence, the remaining 13 bits are used to address the sine wave between 0° and 90°. As the ROM has 256 values, the 8 MSB bits of output phase word $R[12:5]$, i.e., 12th bit to 5th bit of the phase word R, are used for addressing the ROM. From (8), we require the value stored at the present location ($\sin A$) as well as the value stored in the successive location ($\sin C$) for the interpolation technique. Fig. 4 shows the control circuitry that accomplishes this goal. A value of $\sin A$, pointed by $R[12:5]$, is obtained in the present cycle. The control circuit adds a 1 to $R[12:5]$ to point to the successive ROM location, and in the next cycle the value of $\sin C$ is obtained. The multiplexer control, EN, is operating at twice the clock frequency, such that as EN = 1, output of the control circuit points to location of $\sin A$ and when EN = 0, it points to location of $\sin C$. As the truncated bits are $2^d = 2^{13-8} = 32$, the phase register points to a particular ROM value for 32 clock cycles for FCW = 1.

For FCW = 1, d is given by five LSB of R; thus, the sample values obtained from the ROM are shifted left by five positions, which is the same as multiplying it by 32, and later truncated according to the size of the DAC. As in Fig. 4 this is implemented using a 5-bit shift register. The shifted ROM value is added to the output of the multiplier that realizes the first part of (9). From Fig. 4, the inputs to the multiplier are $\sin(C) - \sin(A)$, obtained from the subtractor and the LSB of output phase word denoted by $R[d-1:0]$.

The sine-phase difference algorithm can be introduced in the architecture to reduce the width of the ROM by

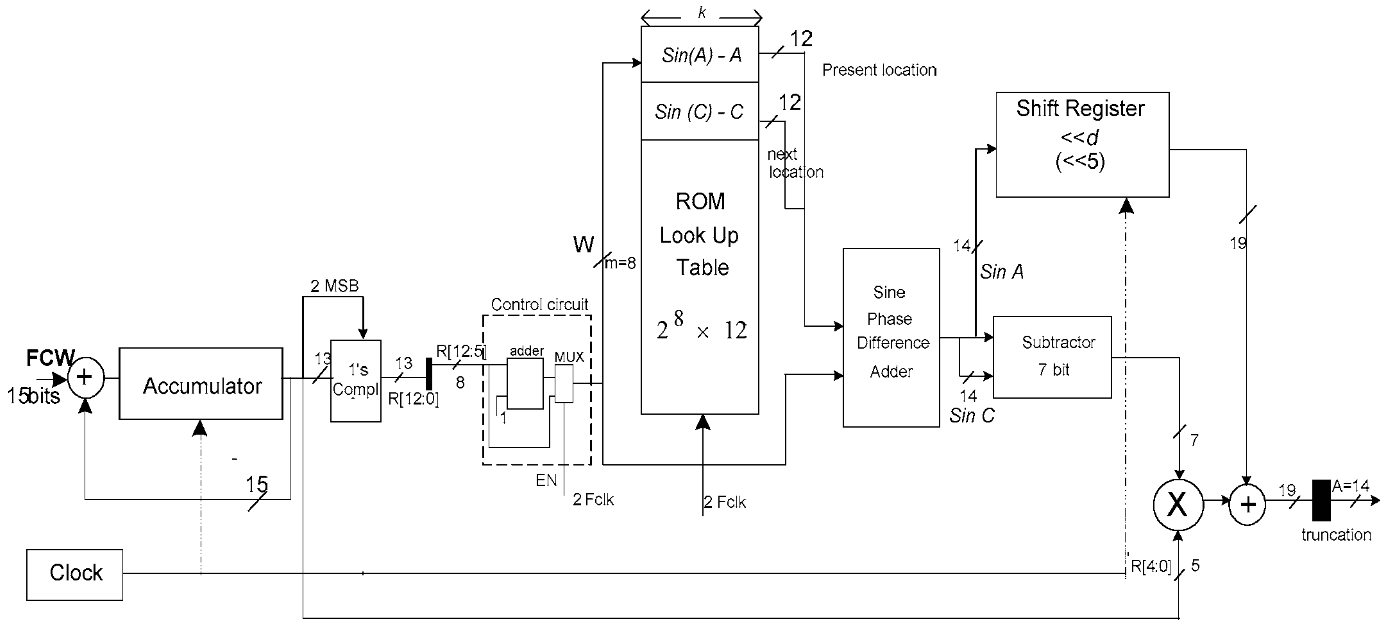

Fig. 3. Phase-to-sine amplitude conversion architecture with linear addressing using a multiplier.

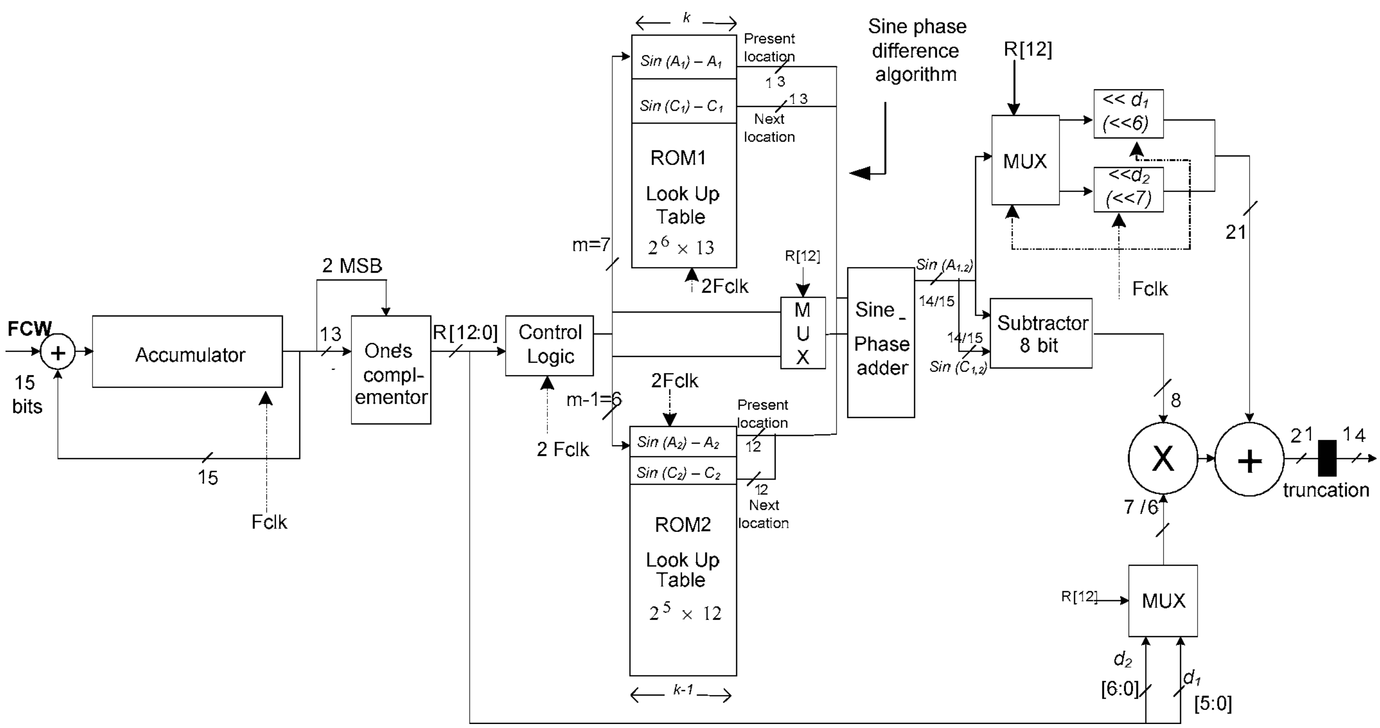

Fig. 4. Phase-to-sine amplitude conversion architecture with nonlinear addressing.

two bits. The values stored in the ROM thus change to $(\sin(x) - x)$ where x is the phase angle of the sinusoid. The inputs to the sine-phase difference adder are the phase given by $W\,(R[m-1:0])$ and the output of the ROM as shown in Fig. 4. Therefore, the length of the output of the block is 14 bits.

The overall sine value is represented with a precision of $14 + 5 = 19$ bits. This value is truncated to 14 bits before being sent to the DAC. The -90 dBc spurious-free dynamic range (SFDR) is obtained for output amplitude resolution of 15 bits. Truncation to 14 bits gives the best spurious response when compared to truncation to either 15 or 13 bits, because in this case the output becomes 15 bits with the MSB sign bit. In order to achieve about the same quantization noise floor due to phase truncation and finite amplitude bits, the number of phase bits is required to be larger than or equal to the number of amplitude bits. So truncation to 14 bits is considered to be optimal. This technique of shifting the ROM value and adding the multiplier value to it to obtain the sine output increases the precision of the output and reduces the complexity of calculations required to obtain the sine values that have to be interpolated.

For the architecture with two ROMs, the sinusoid is partitioned into two parts; 0°–45° and 45°–90°, as shown in Fig. 3. The sine values for phase between 0° and 45° are stored in ROM1, and 45°–90° in ROM2. The phase accumulator is taken to be 15 bits long. The width of the ROMs are taken to be 14 bits and 13 bits, respectively, but due to the architecture having the sine phase difference algorithm incorporated in it, the ROM sizes are given as ROM1 equal to $2^6 \times 12$ bits and ROM2 equal to $2^5 \times 11$ bits. The explanation of ROM2 width being less than ROM1 width is given in Section II-A. The mentioned ROM sizes require the number of samples interpolated between successive stored values to be 64 between 0° and 45° and 128 between 45° and 90° if FCW = 1. The address lines required to address ROM1 are 7 bits, given by $R[12:6]$, and those required for ROM2 are denoted by 6 bits, $R[12:7]$.

The sine phase difference algorithm is incorporated using a single adder. The inputs to the adder are obtained from the phase bits addressing ROM1 or ROM2 at any clock cycle and ROM output that is any one of $\sin(A_1) - A_1$, $\sin(A_2) - A_2$, $\sin(C_1) - C_1$, or $\sin(C_2) - C_2$ at any clock cycle. The phase bits addressing ROM1 or ROM2 are selected using a multiplexer enabled by the $(N-2)$th-bit denoted as $R[12]$.

The output of the subtractor is given by $\sin(C_{1,2}) - \sin(A_{1,2})$. In accordance with (9) and (10), the values of ROM1 are shifted left by $d_1 = 6$ bits and the values of ROM2 are shifted by $d_2 = 7$ bits before being added to the multiplier output to obtain the sinusoid sample. The multiplier input is given by $R[d_1 - 1 : 0] = R[5:0]$ or $R[d_2 - 1 : 0] = R[6:0]$ for ROM1 and ROM2, respectively, and the corresponding values of $\sin(C_{1,2}) - \sin(A_{1,2})$. The whole sine value is represented with a precision of $15 + 6 = 21$ bits. This value is truncated to 14 bits before being sent to the DAC.

The proposed technique requires additional hardware of a 7×8-bit multiplier, shift registers, and a subtractor. Thus the increase in the hardware overhead is the trade-off for the great reduction in the memory requirement.

The sinusoid can further be partitioned to 0°–30°, 30°–60°, and 60°–90°, instead of 0°–45° and 45°–90°, to obtain further reduction in the memory requirement, but for this paper we have only tried partitioning the ROM into two, i.e., 0°–45° and 45°–90°. The values stored in the ROM are optimized to minimize the mean square error by computer simulations. Worst-case spur amplitude of about −88 dBc is achieved with this method. But the simulation results show that a ROM of $2^6 \times 13 + 2^5 \times 12$ bits provides a spurious rejection of −90.3 dBc, which provides the best compression ratio.

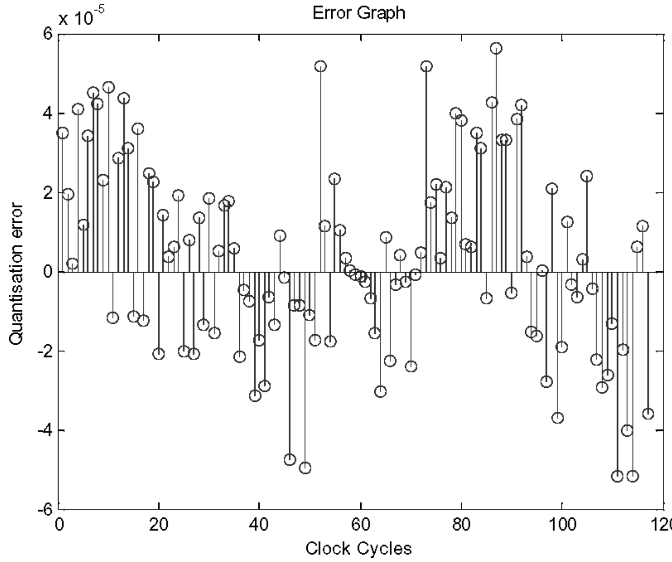

Fig. 5. Error versus samples for = 15 bit, ROM = 28×12 bit, and FCW = 69; error of the order of 10^{-5}.

III. Simulation Results

A computer program was written in MATLAB (The MathWorks, Inc., Natick, MA) to simulate the proposed architecture. The results henceforth are mentioned for both the nonlinear and the linear addressing architectures. The accumulator length and hence the number of output phase word bits are chosen to be $P = 15$ bits. According to [4], an architecture having uncompressed memory with symmetry considerations, having the above-mentioned phase bits, should have a worst-case spur of −90 dBc, which is obtained from the simulations as well. For the proposed linear addressing architecture with multiplier, ROM size is taken to be $2^8 \times 12$ (3072 bits). The worst-case spur thus obtained in this case is also −90 dBc. For nonlinear addressing, the ROM size used is $2^6 \times 13$ for ROM1 and $2^5 \times 12$ for ROM2, adding up to 1216 bits, thus saving 1856 bits as compared to the linear addressing scheme. As shown in Figs. 5 and 6, the error obtained is of the order of 10^{-5}, which yields a worst-case spur of −90 dBc. The presented work has 15-bit accumulator length, and thus has no phase truncation. With increased accumulator length, higher frequency resolution is obtained along with higher signal-to-quantization noise power due to oversampling. The ROM samples have been optimized to reduce the mean square error thus used to obtain the worst-case spur.

The graph in Fig. 7 shows the variation in the spurious response for different widths for ROM1 and ROM2. The widths or word lengths of 13 and 12 for ROM1 and ROM2, respectively, resulted in a magnitude of worst-case spur of −95 dBc. To attain the performance goal of −90 dBc of spurious rejection, the word lengths of 12 and 11 bits for ROM1 and ROM2 are optimal, which result in a worst-case spur of −88 dBc. From the graph it can be observed that, as the word length of the ROMs is reduced, the per-

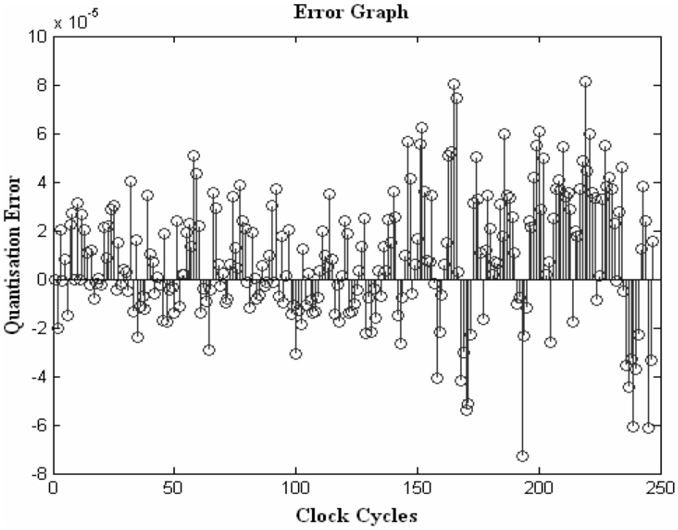

Fig. 6. Error versus samples for = 15 bit, nonlinear addressing, and FCW = 33.

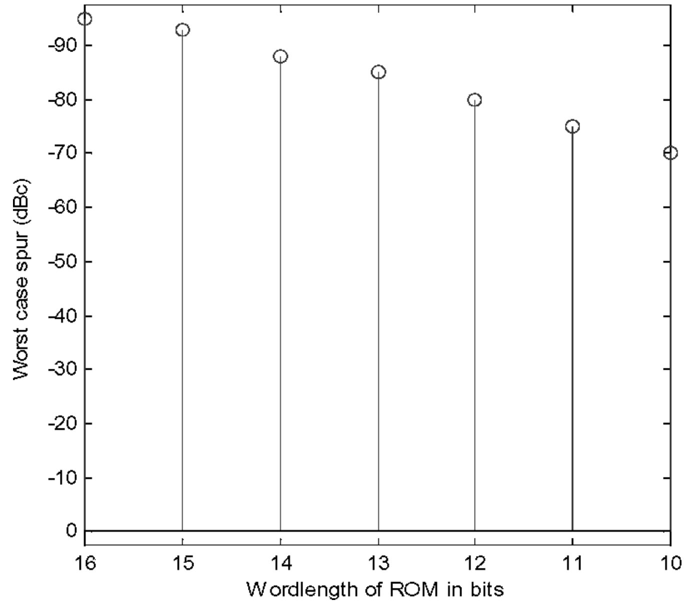

Fig. 8. Memory word lengths versus worst-case spur for linear addressing.

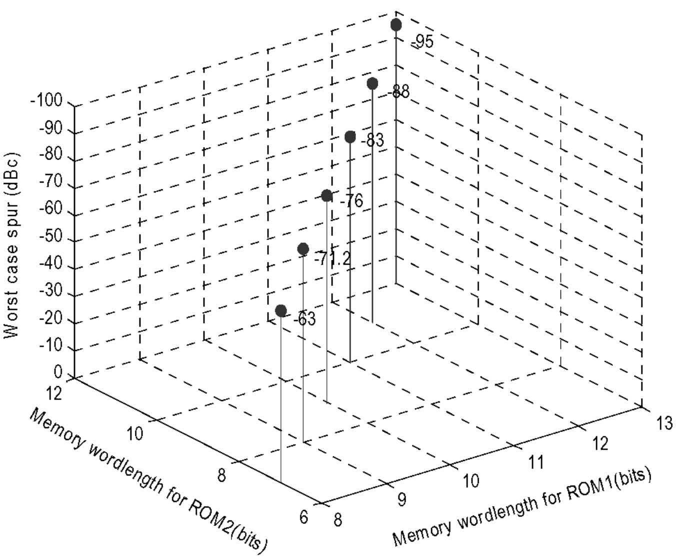

Fig. 7. Memory word lengths versus worst-case spur for nonlinear addressing.

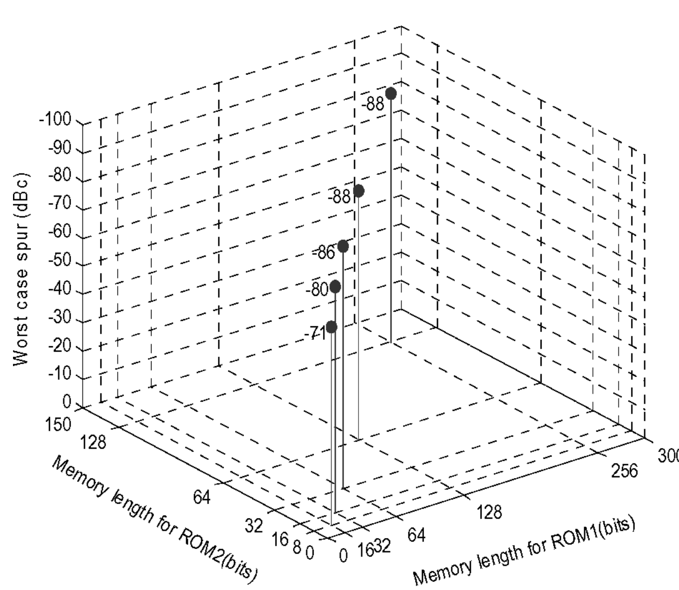

Fig. 9. Memory lengths versus worst-case spur for nonlinear addressing.

formance is degraded. This is caused due to the finite resolution of the ROM leading to quantization error. The objective of attaining -90 dBc of worst-case spur is obtained with word lengths of 12 and 11 bits for ROM1 and ROM2, respectively. Hence the algorithm has been simulated keeping the width of the ROMs constant at these values for different lengths of ROM1 and ROM2, as shown in Fig. 9. As shown in Fig. 8 for linear adressing scheme, the memory wordlength of 14-bits with a memory length of 256-bits achieves the goal of approximately -90 dBc spurious response.

From the graph in Fig. 9 it can be observed that memory lengths of 2^7 and 2^6 for ROM1 and ROM2 meet the requirement of -90 dBc of spurious rejection. It can be observed that even though the memory lengths are increased to 2^8 and 2^7 for ROM1 and ROM2, respectively, there is not much improvement in the performance. One reason for this might be the constant word length of the ROMs, which limits the performance. On the other hand when the memory lengths are decreased to 2^6 and 2^5 for ROM1 and ROM2, respectively, we get a -86 dBc of spurious rejection, which is not much reduction in performance but doubles the compression ratio to 102.4:1. The word length is an important limitation on performance. Hence, by increasing the word length to 13 and 12 bits, respectively, for ROM1 and ROM2 with memory lengths 2^6 and 2^5, respectively, a spurious rejection of -90.6 dBc can be achieved which gives a compression ratio of 94.3:1. This memory dimension is taken as standard for comparison and con-

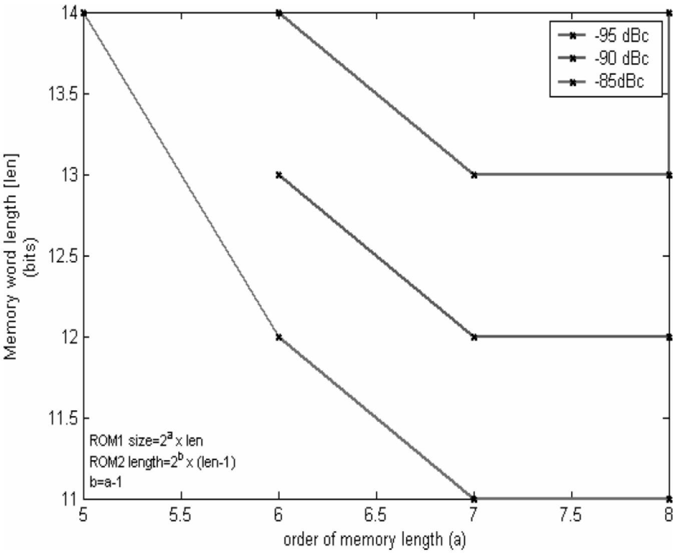

Fig. 10. Different word lengths and memory lengths of ROM1 and ROM2 with the same spurious response.

sidered to be the optimal. The graph in Fig. 10 indicates the possible combinations of ROM1 and ROM2 dimensions giving the same performance. The graph also shows that increasing the memory word length beyond a certain size will not result in further improvement in performance.

IV. Comparison to Prior Art Architectures

In Table I, we compare our proposed DDS ROM compression algorithm to various prior art algorithms. Various algorithms have been developed for compression of look-up tables. A simple architecture for ROM compression, first proposed by Hutchinson [8], works on the angular decomposition technique [9]. This technique uses (11) and thus requires the slicing of the phase angle into constituent angles C and F. The pre-calculated values of the terms stored in separate ROMs are used to obtain the final output.

$$\sin x = \sin(C + F) \cong \sin C + \sin F \times \cos C \tag{11}$$

This was further improved by Sunderland *et al.* [10] using three slices. But, as shown in Table I, both architectures have achieved much less compression ratio. The architecture proposed by Nicholas *et al.* [6], [7] consists of a coarse and a fine ROM structure. The values stored in the coarse ROM are given by $\sin(A + B)$ and those in fine ROM are $(\sin(C)\cos(A))$. The stored values minimize the mean square error component such that the noise power at the output is reduced. Storage bits were further reduced by selecting an optimum segmentation scheme. Sine amplitude compression named the sine-phase difference algorithm was also used to reduce storage bits. This architecture used to be very popular due to its simplicity and minimum hardware requirements. In the Nicholas architecture, more values are indirectly stored.

Curticapean and Niittylahti [11] have an architecture similar to that in [8] but have used three ROMs. The values stored are $\sin(C)$, $\sin(F)$, and $\cos(C)$, with the terms multiplied externally. The reported 455:1 compression ratio is obtained by comparison with uncompressed memory. When the proposed architecture is compared with quarter sine wave symmetry memory, the compression ratio obtained is 94.3:1. When it's compared with uncompressed memory, the obtained compression ratio is 404:1, as specified in Table I.

The architecture proposed by Bellaouar *et al.* [12] uses the first two terms of the Taylor series expansion for a sinusoid. This kind of architecture is categorized as the first-order polynomial approximation-type ROM compression technique. The two ROMs store the values of a constant first term and the second term having the slope coefficient of the sine wave.

El Said and Elmasry [13] performed the linear interpolation from the center of each linear segment unlike that presented in [12]. There is a 50% reduction in the ROM size by performing interpolation around a value in the middle of the interpolation range. However, their compression ratio degrades for large ROM size. Therefore, in other words, a large accumulator length causes reduction in the compression ratio, thereby creating a trade-off between ROM size and the frequency resolution.

Langlois and Al Khalili [9], [14] have proposed a technique where the sine wave is divided into linear segments that are defined such that the output noise sequence samples have little correlation with each other, flattening the noise spectrum and increasing the SFDR. Also, they have made extensive use of multiplexers to remove restrictions on systems having multipliers, similar to the architecture proposed by Liu *et al.* [15]. The consequence, however, is that the SFDR will be degraded for FCW values that do not force all output samples to be cycled through, or if very short sequences of oscillations are synthesized prior to changing the FCW. But the output resolution of this architecture is smaller than that achieved in this work. With the increase of the output resolution, their error increases, but the SFDR is limited by the number of input bits to the DAC due to DAC quantization noise.

Liu *et al.* [15] have proposed a multiplierless technique for ROM compression that utilizes a linear interpolation technique with unequal length of segments. This is implemented in a way such that the slope includes a wide range of powers-of-two. Thus circuit complexity is greatly increased.

As the speed of technology is increasing, calculation of values is faster as well as easier; hence the memory requirements can be reduced by calculating the values that have to be interpolated rather than storing them. This is the basis for the proposed architecture where all of the values that have to be interpolated are calculated, thereby greatly reducing the memory requirements. Further, the number of values to be calculated can be increased, which reduces the number of values that have to be stored, by exploiting the properties of sinusoid.

TABLE I

Comparison of Proposed DDS ROM Compression Algorithm to Prior Art Algorithms.

Method	Phase (P) & amplitude (A) resolution	Needed ROM (bits)	Ref. look-up table	Total compress. ratio	Spectral purity	Comments
Hutchinson's technique [8]	P = 12 A = 8	$2^7 \times 6$ $2^{10} \times 2$	$2^{12} \times 8$	11.6:1*	−72.245	(3,4,3) phase segmentation, two adders required
Sunderland *et al.* architecture [10]	P = 12 A = 8	$2^7 \times 8$ $2^6 \times 3$	$2^{12} \times 8$	27:1*	−72.245	(3,3,4) phase segmentation, two adders required
Nicholas *et al.* architecture [6], [7]	P = 15	$2^8 \times 9$ $2^8 \times 4$	$2^{13} \times 14$	34.4:1	−90	(4,4,5) phase segmentation, two adders required
Bellaouar *et al.* architecture [12]	P = 12 A = 8	$2^5 \times 8$ $2^5 \times 5$	$2^{12} \times 8$	78.8:1*	−72.245	(5,5) phase segmentation, a 5×5 multiplier, and two adders
El Said and Elmasry architecture [13]	P = 12 A = 8	$2^4 \times 8$ $2^4 \times 8$	$2^{12} \times 8$	128:1*	−72.1	(4,6) phase segmentation, 8×7 multiplier, two adders
Langlois and Al Khalili [9], [14]	P = 18 A = 14	$2^6 \times 15$	$2^{14} \times 16$	273:1	−96	64 linear segmentation, with multiplexer
Curticapean and Niittylahti [11]	P = 16 A = 16	15×2^6 14×2^6 7×2^6	$2^{16} \times 16$	455:1*	−96	5×7 bits two multipliers and two adders
Proposed architecture with linear addressing	P = 15 A = 15	$2^8 \times 12$	$2^{13} \times 14$ $2^{15} \times 15$	37.3:1 160:1*	−89	5×7 multiplier, two adders
Proposed architecture with nonlinear addressing	P = 15 A = 15	$2^6 \times 13$ $2^5 \times 12$	$2^{13} \times 14$ $2^{15} \times 15$	94.3:1 404:1*	−90.3	6×7 multiplier, two adders

*Represents the compression ratio measured with reference to the uncompressed memory instead of the otherwise used reference of quarter-sine-wave symmetry memory. The sine-wave symmetry memory = a quarter of uncompressed memory.

The inherent property of our architecture is that the compression ratio greatly increases for increasing size of the phase accumulator. Compared to prior art techniques, the proposed architecture can have a better compression ratio for a larger size of phase accumulator and better output amplitude resolution so that the SFDR is not limited by the DAC quantization noise.

For P = 15 bits, the complexity of the proposed architecture is around 8300 gates. The comparison of our architecture to that of Nicholas , . [6], [7] for P = 15 bits is given in Table I, showing much better compression ratio with the same spurious performance.

V. Implementation and Measurements

We have coded the linear, nonlinear, and traditional DDS architectures using Verilog RTL and then implemented them using a Xilinx Spartan II FPGA. Due to the PCB board noise at about −30 dBc, we can use only an 8-bit DAC for our implementation. We therefore modified our architecture to have a 15-bit accumulator that is truncated to 9 phase bits and a compressed ROM that generates an 8-bit sinusoidal waveform. Choosing 9 ROM phase bits will help fully utilize the number of DAC bits and meanwhile can still demonstrate the ROM compression algorithm. The 8-bit DAC has an effective number of 5–6 bits that results in about 30 dBc measured SFDR. The clock frequency applied is 25 MHz. For FCW = 1, the characteristic equation for DDS gives the output frequency as

$$f_{clk} = 25 \text{ MHz}$$
$$f_{out} = \frac{25\ M}{2^{15}} \times (1) = 762\ .93 \text{ Hz}. \tag{12}$$

For the nonlinear architecture measurement plot, as shown in Fig. 11, the ROM sizes are given by ROM1 ($2^3 \times 8$), ROM2 ($2^2 \times 7$); 4 bits of the phase word are used to address ROM1 and 3 bits are used to address ROM 2, which has widths of 8 and 7 bits, respectively. The compression ratio achieved in this case is 11.13:1 when compared with the traditional architecture.

Fig. 11 shows the output spectrum of the DDS with nonlinear architecture with FCW = 1, phase accumulator = 15 bits, DAC resolution = 8 bits, and clock frequency = 25 MHz. The phase truncation from 15 bits to 9 bits contributes harmonic spurs in the output spectrum. This is because the truncation error is periodic [4], [8], [16]. The power of the unwanted harmonics in the output can be significantly reduced by eliminating the periodicity of truncation error by introducing techniques such as sigma delta modulation. For an N-bit accumulator output truncated into P-bit phase, the phase truncation spurs are mixed with the DDS output frequency, generating spurs at offset frequencies of [6]

$$f_{spur} = F_n \cdot \frac{GCD\left(FCW, 2^{N-P}\right)}{2^{N-P}} \cdot f_{clk}, \tag{13}$$

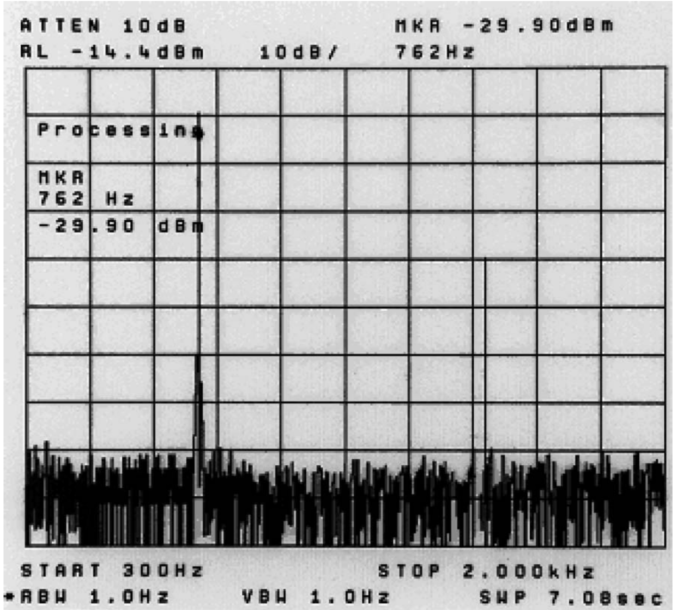

Fig. 11. Spectrum of DDS with nonlinear architecture for FCW = 1; phase accumulator = 15 bits, DAC resolution = 8 bits, clock freq. = 25 MHz.

where F_n is an integer to sequentially number the spurs and $GCD(a, b)$ denotes the greatest-common-divisor of a and b.

VI. Conclusion

A novel architecture that exploits the slope of the sine function for compression of storage requirement has been proposed. The architecture has been compared with the prior art architectures in simulations and measurements. The simulated and measured results demonstrated that the same spurious performance can be achieved with a much better compression ratio compared to existing DDS architectures.

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Lakshmi Chimakurthy was born in Chirala, Andhra Pradesh, India, in 1981. She received her B-TECH degree from Jawaharlal Nehru Technological University, Hyderabad, India, in 2002. She has been working towards her M.S. degree as a Research Assistant at Alabama Microelectronics Science and Technology Center, Auburn University, Auburn, AL, since January 2003. Her research interests include high speed DDS design, radio frequency IC design, analog and mixed signal circuit design, and wireless communication. Currently, she is with Texas Instruments, Dallas, TX.

Malinky Ghosh was born in Bellary, Karnataka, India, in 1980. She received her Bachelors degree from Nagpur University, Nagpur, India, in 2002. She has been working towards her M.S. degree as a Research Assistant at Alabama Microelectronics Science and Technology Center, Auburn University, Auburn, AL, since January 2003. Her research interests include sigma delta modulators, radio frequency IC design, analog and mixed signal circuit design, and wireless communication. From August 2004 until December 2004, she was with Qualcomm, San Diego, CA, as an intern. Since June 2005 she has been with Texas Instruments, Tucson, AZ, as an intern.

Fa Foster Dai (M'92–SM'00) received the B.S. degree in physics from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1983. He received a Ph.D. degree in electrical and computer engineering from Auburn University, Auburn, AL, in 1997 and another Ph.D. degree in electrical engineering from The Pennsylvania State University, University Park, PA, in 1998.

From 1986 to 1989, he was a Lecturer at the UESTC. From 1989 to 1993, he was with the Technical University of Hamburg, Germany, working on microwave theory and RF designs. From 1997 to 2000, he was with Hughes Network Systems of Hughes Electronics, Germantown, MD, where he was a Member of Technical Staff in very large scale integration (VLSI) engineering, designing analog and digital ICs for wireless and satellite communications. From 2000 to 2001, he was with YAFO Networks, Hanover, MD, where he was a Technical Manager and a Principal Engineer in VLSI designs, leading high-speed SiGe IC designs for fiber communications. From 2001 to 2002, he was with Cognio Inc., Gaithersburg, MD, designing radio frequency ICs (RFICs) for integrated multi-band wireless transceivers. From 2002 to 2004, he was a RFIC consultant for Cognio Inc. In August 2002, he joined the faculty of Auburn University, where he is currently an Associate Professor in electrical and computer engineering. His research interests include VLSI circuits for digital, analog and mixed-signal applications, RFIC designs for wireless and broadband communications, ultra-high frequency synthesis, and analog and mixed signal built-in self-test (BIST). He is the co-author of *Integrated Circuit Design for High-Speed Frequency Synthesis*, Artech House Publishers, Feb, 2006.

Dr. Dai has served as a Guest Editor for the *IEEE Transactions on Industrial Electronics.* He currently serves on the Technical Program Committee of the Symposium on VLSI Circuits.

Richard C. Jaeger (F'86) was born in New York City on September 2, 1944. He received the B.S. and M.E. degrees in Electrical Engineering in 1966 and the Ph.D. degree in 1969, all from the University of Florida, Gainesville, FL. From 1969 to 1979 he was with the IBM Corporation working on precision analog design, integrated injection logic, microprocessor architecture and low temperature MOS device and circuit behavior. In 1979 he joined Auburn University, Auburn, AL, where he is Distinguished University Professor in Electrical and Computer Engineering and Associate Director of the Wireless Engineering Program. He has published over 200 technical papers and articles, the Addison Wesley book *Introduction to Microelectronic Fabrication (2E)*, and the McGraw-Hill text *Microelectronic Circuit Design (2E)*.

Dr. Jaeger is a past editor of the *IEEE Journal of Solid-State Circuits* and is now President of the IEEE Solid-State Circuits Society. He was Program Chairman for the 1993 International Solid-State Circuits Conference, and Chairman of the 1990 VLSI Circuits Symposium. From 1980 to 1982 he served as founding Editor-in-Chief of *IEEE MICRO*, and subsequently received an Outstanding Contribution Award from the IEEE Computer Society for development of that magazine.

Dr. Jaeger was elected Fellow of the IEEE in 1986 and was appointed to the Distinguished University Professorship by Auburn University in 1990. He received the 1998 McGraw-Hill/Jacob Millman Award from the IEEE Education Society and the 2004 IEEE Undergraduate Teaching Award. In his spare time, he is an avid amateur radio operator (K4IQJ), enjoying dx and contest operation.